Syllabus Year: 2024-2025

CHRIST (Deemed to be University), Bengaluru - 560029

End Semester Examinations - November 2025

Bachelor of Computer Applications SEMESTER - I

Course Code: BCA101-1 **Max Marks: 100**

Course Name: FOUNDATIONAL MATHEMATICS **Duration: 03:00**

General Instructions:

- All rough work should be done in the answer script. Do not write or scribble in the question paper except your register number.
- Verify the Course code / course title and number of pages of questions in the question paper.
- Make sure your mobile phone is switched off and placed at the designated place.
- Malpractices will be viewed very seriously.
- Answers should be written on both sides of the paper in the answer book. No sheets should be detached from the answer book.
- Answers without the question numbers clearly indicated will not be valued. No page should be left blank in the middle of the answer book.

Course Outcome: Students will be able to:

CO1: Reformulate statements from common language to formal logic using the rules of propositional and predicate calculus and assess the validity of arguments.

CO2: formulate and interpret statements presented and determine their validity by applying the rules and methods of propositional logic.

CO3: construct elementary proofs using ordinary and strong induction in the context of studying the properties of recursion.

CO4: apply basic counting principles including the pigeonhole principle and rules for counting permutations and combinations.

CO5: apply the logical structure of proofs and work symbolically with connectives and quantifiers to produce logically valid, correct and clear arguments

Section A **3*8=24**

1)	Analyze the statement 'For every even number n, n+2 is also an even number' using propositional calculus and determine its validity.	**3**	(CO1, L4)
2) a)	Analyze the statement 'For every prime number p greater than 2, p+2 is also a prime number' by constructing a truth table for its symbolic representation.	**3**	(CO5, L4)
	OR		
b)	Examine and differentiate between a symmetric relation and an antisymmetric relation using examples on finite sets.	**3**	(CO5, L4)
3)	Demonstrate that for any real number x, if x is an integer, then the sum of x and 2 is also an integer.	**3**	(CO5, L3)
4)	Discuss the concept of an equivalence class using a specific example and explain how it relates to the properties of an equivalence relation.	**3**	(CO3, L2)
5) a)	Discuss the concept of a valid argument in formal logic by constructing a truth table for the following statement: 'If it is raining, then the ground is wet' and interpreting its validity.	**3**	(CO1, L2)
	OR		
b)	Describe the process of determining whether a given argument is valid using predicate calculus. Provide an example to illustrate your explanation.	**3**	(CO1, L2)

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6)	Demonstrate using formal logic that for every positive integer n, if n is odd then 2n + 1 is also an odd number.	**3**	(CO1, L3)
7) a)	Analyze the statement 'For every odd positive integer n, n^2 + 1 is also an even number.' and provide a proof or counterexample.	**3**	(CO3, L4)
	OR		
b)	Examine and differentiate between a symmetric matrix and an anti-symmetric matrix using examples from 2x2 matrices. Also, prove that any square matrix can be expressed as the sum of a symmetric and an anti-symmetric matrix.	**3**	(CO3, L4)
8)	Examine and explain why the statement 'For every real number x, if x is rational then x^2 is also rational' is true or false using formal logic and provide a proof.	**3**	(CO5, L4)
	Section B		**8*7=56**
9)	A relation R on a set A = {a, b, c, d} is defined as follows: R = {(a, a), (b, b), (c, c), (d, d), (a, b), (b, a), (b, c), (c, b)}. Determine whether the relation R is symmetric. If it is not symmetric, modify R to create a new relation S that is symmetric and demonstrate using formal logic that S satisfies the properties of symmetry. Additionally, represent both relations R and S as matrices.	**8**	(CO1, L3)
10) a)	Demonstrate that among any group of five distinct positive integers less than or equal to 10, there must be at least one pair whose difference is divisible by 3. Then, construct an example to show that this result does not hold for four such integers.	**8**	(CO4, L3)
	OR		
b)	Construct a set A containing exactly 8 elements and define a relation R on A such that R contains at least 12 ordered pairs and demonstrate whether R is reflexive, symmetric, or transitive using a relation matrix. Additionally, apply the pigeonhole principle to explain why your constructed relation must have at least one pair (a, b) where aRb implies bRa, under certain conditions.	**8**	(CO4, L3)
11)	A library has 10 different books on mathematics and 10 different books on computer science. A student wants to select a collection of these books for her study. (i) In how many ways can she choose 3 books if at least one book must be from each subject? (ii) If the student decides to borrow exactly 4 books, in how many ways can this be done such that no more than two books are borrowed from any single subject?	**8**	(CO4, L2)
12) a)	Construct an elementary proof using strong induction to show that for any positive integer n, the number 7 divides at least one of the numbers in the sequence defined by a_1 = 2, and a_{n+1} = 3a_n + 4 for all n ≥ 1. Demonstrate your understanding by providing the base case, induction hypothesis, and inductive step clearly.	**8**	(CO3, L3)
	OR		
b)	Construct a set A containing exactly 8 elements such that it forms an equivalence class under the relation R defined as follows: For any two elements x, y in A, (x, y) R if and only if x - y is divisible by 3. Demonstrate your construction using both set notation and a table showing all ordered pairs in R. Additionally, prove that R is an equivalence relation on the set A.	**8**	(CO3, L3)
13)	Discuss using a valid argument in formal logic why every odd positive integer can be expressed as 2k + 1 for some non-negative integer k, and then construct an elementary proof by strong induction to show that the sum of the first n odd positive integers is always a perfect square.	**8**	(CO3, L2)

14) a)	In a university department with 12 professors, each professor specializes in one of four subjects: Mathematics, Physics, Chemistry, or Biology. (i) Describe how many ways can you select a committee of 4 professors if at least two must specialize in different subjects? (ii) Explain the principle used to calculate the number of ways to distribute 10 distinct research papers among these 12 professors such that no professor receives more than one paper.	**8**	(CO4, L2)
	OR		
b)	Discuss using a valid argument in formal logic why for any set A with more than 4 elements, there must exist at least one subset of A containing exactly 3 elements such that their sum is even. Use the pigeonhole principle and concepts from permutations and combinations to explain your reasoning.	**8**	(CO4, L2)
15)	Analyze how many ways can 10 letters be distributed into 5 distinct mailboxes such that each mailbox receives at least one letter. Also, determine the number of ways to distribute these letters if no two consecutive letters are placed in the same mailbox. Provide a detailed explanation using appropriate counting principles and permutations.	**8**	(CO4, L4)
	Section C		**10*2=20**
16)	Examine and analyze how the principles of propositional logic can be used to verify the validity of an argument in a real-world scenario involving conditional statements. Specifically, consider the following situation: A university library decides to implement a new policy where if a book is requested by at least 3 students on any given day (denoted as P), then it will be made available for immediate checkout (denoted as Q). Formulate this policy into a logical statement and use truth tables to determine under what conditions the policy will hold true. Discuss how this analysis can assist in decision-making processes within the library.	**10**	(CO2, L4)
17) a)	Determine whether the following statement is true or false: 'For every positive integer n, there exists a unique positive integer m such that 2m - 1 = n'. If the statement is true, provide a constructive proof to demonstrate its validity by formulating and interpreting logical statements. If the statement is false, construct a counterexample to disprove it.	**10**	(CO2, L3)
	OR		
b)	Determine and construct an equivalence relation R on a finite set A = {1, 2, 3, 4, 5} such that there are exactly two distinct equivalence classes. Also, represent this relation using a directed graph. (i) Write down the set A and explicitly define the relation R by listing all ordered pairs in R. (ii) Draw the directed graph of the relation R to visually demonstrate its properties. (iii) Explain why your defined relation satisfies the conditions of an equivalence relation.	**10**	(CO2, L3)

Revised Bloom's Taxonomy (RBT) Levels

L1 - Remember
L2 - Understand
L3 - Apply
L4 - Analyse
L5 - Evaluate
L6 - Create

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